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ORIGINAL PAPER

HIM-STRAT: a neural network-based model for snow cover simulation and avalanche hazard prediction over North-West Himalaya

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Abstract
Artificial neural network (ANN)-based models have been developed for simulation of snowpack parameters—RAM hardness, shear strength, temperature, density, thickness and settlement of snowpack layers using manually observed weather data. The simulated snowpack parameters have been used for development of ANN for avalanche prediction. The complete scheme of simulation of snowpack parameters and avalanche prediction has been named as HIM-STRAT and developed for Chowkibal–Tangdhar region in North-West Himalaya using weather and snow stratigraphy data collected at a representative observatory in that region. Weather and snowpack data collected during the period from 1992 to 2016 have been analysed to generate the database of prominent snowpack layers and associated weather variables. Snowpack parameters have been simulated using randomly selected 490 (80%) data points and tested with 123 (20%) data points. Simulated snowpack parameters—RAM hardness, shear strength of the weakest snowpack layer and overburden pressure on the weakest layer—have been used to derive stability index of snowpack (SIS). The SIS and other snowpack parameters such as snowpack height, RAM hardness, shear strength, storm snow and snow temperature have been derived for past 22 winters (1992–2014) and used for prediction of avalanches. The HIM-STRAT has been validated through computation of root-mean-square error of simulated snowpack parameters and accuracy, bias, false alarm rate and Heidke Skill Score of avalanche prediction for five winters from 2014 to 2019. The performance of HIM-STRAT for simulation of snowpack parameters and prediction of avalanches has been found reasonably good and discussed in detail.

Keywords Snowpack simulation · Snow stratigraphy · Snowpack parameters

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1 Introduction

Snow cover simulation over space and time has been found useful in many applications such as weather, hydrology, glaciology and avalanche forecasting. Seasonal snow cover in the Cryosphere is a very significant climate forcing (Flanner et al. 2011) with significant influence on energy budget of the soil and the atmosphere. Simulation of snow cover for snowpack stability assessment is the key to avalanche forecasting. The detailed information about snowpack stability is collected through a manual experiment in the field called snowpack stratigraphy. It is quite time-consuming, sometime dangerous and, due to highly spatially variable nature of mountainous snowpack, cannot provide true picture of snowpack stability over larger areas. Therefore, snowpack parameters need to be simulated in space and time for snowpack stability assessment over larger areas and avalanche hazard prediction.

Simulation of snowpack parameters worldwide is based on physical modelling approach. Development of physical models for snow cover processes started with the works of Colbeck (1972), Navarre (1975), Obled and Rosse (1977) and Anderson (1976). They used weather conditions for simulation of evolution of snow cover. The snow cover models such as SNOTHERM (Jordan 1991) and SAFRON-CROCUS-MEPRA model chain (Brun et al. 1989, 1992) were developed and further improved by incorporating more physical processes such as snow metamorphism and absorption of solar radiation. In these models, snowpack behaviour has been modelled based on simplified mixture theories of Morland et al. (1990). Simplifying assumptions such as a rigid ice matrix (no settling) (Gray and Morland 1994), no water content (dry snowpacks) (Gray and Morland 1995) or no snow metamorphism (Bader and Weilenmann 1992) have been used to derive these models. Snow cover model—SNOWPACK—has been developed for simulation and temporal evolution of snow stratigraphy by including complex physical processes through numerical modelling and using data from automated weather stations (Bartelt and Lehning 2002; Lehning et al. 1999, 2002a, b) and numerical weather prediction (NWP) models (Iwamoto et al. 2008; Bellaire et al. 2011, 2013).

In recent years, mainly two advanced snow cover models such as French model chain SAFRAN-CROCUS-MEPRA (SCM) (Durand et al. 1999; Lafaysse et al. 2013; Vionnet et al. 2012) and Swiss snow cover model—SNOWPACK—have been used for various applications including avalanche forecasting (e.g., Durand et al. 1999; Schirmer et al. 2010). Both SNOWPACK and Crocus make use of automatic weather station data for simulation of snow cover processes (Mitterer and Schweizer 2013; Vionnet et al. 2016; Bellaire et al. 2017; Quéno et al. 2016, 2018). These models are also able to forecast the evolution of snow cover using numerical weather prediction data (Bellaire and Jamieson 2013a; Quéno et al. 2016; Vernay et al. 2015; Vionnet et al. 2016). These models have been validated mainly for snowpack height and stratigraphy (e.g., Bellaire and Jamieson 2013a; Bellaire et al. 2011; Schirmer and Jamieson 2015; Vionnet et al. 2012) and rarely for snowpack stability (e.g., Bellaire and Jamieson 2013b; Vernay et al. 2015).

The physical snow cover models such as SNOWPACK and SCM are able to simulate snowpack parameters using hourly data of either automatic weather station (AWS) or high-resolution numerical weather prediction (NWP) models. In Indian Himalaya, the physical model—SNOWPACK—could be run only for limited regions due to very thin AWS network and discontinuous data during snow storms. Moreover, the bias-corrected NWP model output at desired spatial and temporal resolution for Himalaya is presently not available at Snow and Avalanche Study Establishment (SASE) to run the physical snow cover models. Though Spreitzhofer and Sperka (2015) used MetGIS technologies for ultra-high-resolution weather predictions for the Himalaya, high-resolution MetGIS data for Himalaya are available only commercially. Thus, the physical models produce incomplete and wrong simulation results with missing hourly data of AWS and could not be run operationally for most of the regions of Himalaya.

Avalanche danger prediction using manually observed snow cover information was first attempted by Schweizer and Fohn (1996) using a commercially available decision making software. This model with the inclusion of knowledge in the form of expert rules gave cross-validation hit rate (POD) up to 70%. It could not run fully automatically due to the requirement of manual input of snow cover information. To automate avalanche forecasting using meteorological variables and snow information, Brabec and Meister (2001) used nearest neighbour method with an accuracy (PC) of only 52%. Low accuracy of this model was mainly attributed to the lack of snow cover information. The French model chain (SCM), the only automated physical model giving level of avalanche risk for operational purpose, modelled avalanche risk with a hit rate (POD) of 75%. Hence, incorporation of snow cover information into avalanche forecasting models is essential for improvement of avalanche forecast.

Numerical simulation of snow cover requires historical database of weather and snowpack parameters. In North-West Himalaya, snow and meteorological and stratigraphy data are collected manually at different meteorological stations of Snow and Avalanche Study Establishment (SASE). At these stations, there is a good record of snow and meteorological data of last more than four decades. Using these data, a few numerical avalanche forecasting models have been developed for different regions in Pir Panjal and Great Himalaya (Naresh and Pant 1999; Singh et al. 2005; Singh and Ganju 2008; Joshi et al. 2010; Joshi and Srivastava 2014; Joshi et al. 2018). All these models make use of daily recorded snow and meteorological data (Class-III) for avalanche prediction. However, snowpack data (Class-II) that provide information about snowpack stability being more relevant to the avalanche formation are more useful for prediction of avalanches. There are three data classes for assessment of snowpack stability based on ease of their interpretation and relevance for snowpack stability assessment (McClung and Scherer 2006). The data of higher class number are more uncertain in assessing the snowpack stability than that of the lower class. Class-I data deal with direct relationship between loads on weak snowpack layers. It includes current avalanches and different snow loading tests. Class-II data are collected from within the snowpack, and they provide an evidence about presence, strength and loading of weak snow layers. It is less directly related to snowpack stability than Class-I data. Class-III data provide an indirect evidence about current and future snowpack stability. They include snow and meteorological data collected at or above the snow surface. Class-II data being more relevant than Class-I data for assessment of snowpack stability and avalanche prediction, a neural network-based snow cover model, HIM-STRAT, has been developed for simulation of snowpack parameters such as RAM hardness, shear strength, density, settlement, temperature and thickness of snowpack layers and prediction of avalanche hazard in Chowkibal–Tangdhar region in North-West Himalaya. This simulation technique can be helpful and computationally cost-effective for spatial and temporal snow cover simulation and associated avalanche hazard prediction worldwide. The simulated seasonal snow cover can also be useful for hydrological and climatological applications.

2 Data

Snow stratigraphy and weather data of past 24 winters (1992–2016) collected at Stage-II, a meteorological station of SASE in J&K, India (Fig. 1), have been used for development of HIM-STRAT. Stage-II observatory (2650 m m.s.l.) is in the eastern side of Chowkibal–Tangdhar (C-T) region and situated near the Chowkibal valley. The C-T region comprises an area of approximately 100 km^2. There are 26 registered avalanche sites along C-T national highway within the stretch of 38 km. Stage-II being the only manual observatory in C-T region is representative of snow, meteorological and avalanche characteristics of the entire region. According to a study by Sharma and Ganju (2000), the entire C-T region including Stage-II observatory falls in Lower Himalayan Climatic zone of the Himalaya. Pir Panjal and Shamshabari mountain ranges of the Himalaya are part of the Lower Himalayan Climatic zone which is characterised by warm temperatures, high precipitation, deep snowpack and short winter period of 3–4 months. The average height of mountains in this climatic zone varies from 2000 to 4000 m. Thus, the climatic conditions of Lower Himalayan Climatic zone are similar to that of maritime snow climate.

In the C-T region, a total of 292 stratigraphies were conducted on different aspects during past 24 winters out of which 186 were carried out on zero aspect (plane ground) used for development of the HIM-STRAT. In each snow stratigraphy, prominent snow layers and corresponding weather variables were identified through manual analysis of the stratigraphy and seasonal snow cover build-up plots. The time series plot of seasonal snow cover build-up provides information about the number of snow layers, time and duration of snow storms in which the layers were formed. The stratigraphy carried out during the period from December to March has only been analysed due to availability of limited stratigraphy data during late winter (April). Moreover, snowpack properties of late winter isothermal snowpack significantly differ from rest of the snowpack data. Analysis of snow stratigraphies is conducted on zero aspect and corresponding snow cover build-up resulted in a database of 613 snow layers. The weather variables recorded during the formation of a snow layer and snowpack parameters of that layer have been taken as complete data corresponding to that layer. Snow and weather data corresponding to a layer involve maximum, minimum and average temperature of snow storm in which the layer was formed, total sun shine hours and days for which the layer had solar exposure, layer thickness, storm snow, snowpack depth, snow height above the layer, snowpack settlement, density, shear strength and RAM resistance of the layer. The prediction variables for one snowpack parameter slightly differ for the

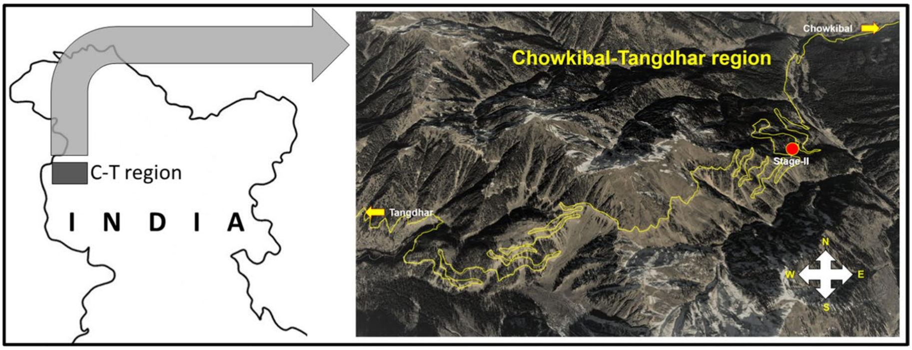

Fig. 1 Location of Stage-II observatory in Chowkibal–Tangdhar region in North-West Himalaya (Google Earth Image)

other. The prediction variables (snowpack parameters and weather variables corresponding to a snow layer) used in the simulation of snowpack parameters are summarised in Table 1.

3 Methodology

Simulation of snowpack parameters and avalanche hazard classification is based on feed-forward artificial neural network. The development steps of the model are shown in the block diagram in Fig. 2. Snowpack stratigraphy data of past 24 winters from 1992 to 2016 have been analysed for selection of prominent snow layers and associated snowpack parameters and weather variables. Out of 613 data points of prominent snow layers, 490 were used for training and 123 for testing of neural networks. The input data used for development of neural network for one of the snowpack parameters differ slightly for the other. Snow and meteorological variables are listed in Table 1, excluding the variable to be simulated serve as the model input for simulation. Thus, five different sets of input variables have been used for simulation of five snowpack parameters.

The simulated snowpack has been compared with the observed stratigraphy through computation of agreement scores for snowpack parameters and overall agreement score of the snowpack. Computation of agreement score is based on the study carried out by Lehning et al. 2001. For comparison with the observed snowpack, the modelled snowpack has been linearly stretched to compensate for snow depth differences. The stretch factor which is constant for all the layers is calculated as follows:

$$\text{Stretch factor (s)} = \text{Observed snow depth}/\text{modelled snow depth}$$

After adjusting the snow depth, the values of observed and simulated snowpack parameters (RAM resistance, snow temperature and density) have been taken at equal intervals for computation of the agreement score for each of the snowpack parameters. For 'N' equal intervals, the agreement score of a snowpack parameter is calculated as follows:

$$\text{Agreement score} = \left(N - \left(\sum_{i=1}^{N} |\text{Observed}\,(i) - \text{Simulated}\,(i)|\right)/ \right.$$
$$\left. \text{Max (Observed, Simulated)} - \text{Min (Observed, Simulated)}\right)/N$$

Table 1 Snow–meteorological and snowpack parameters used in the simulation

Parameters	Units
1. Maximum temperature during snow storm	°C
2. Minimum temperature during snow storm	°C
3. Average wind speed during snow storm	km/h
4. Storm snow	cm
5. Standing snow	cm
6. Snow temperature	°C
7. Total sun shine hours	h
8. RAM resistance	kg
9. Shear strength	kg
10. Density	g/cm^3
11. Snowpack settlement	cm

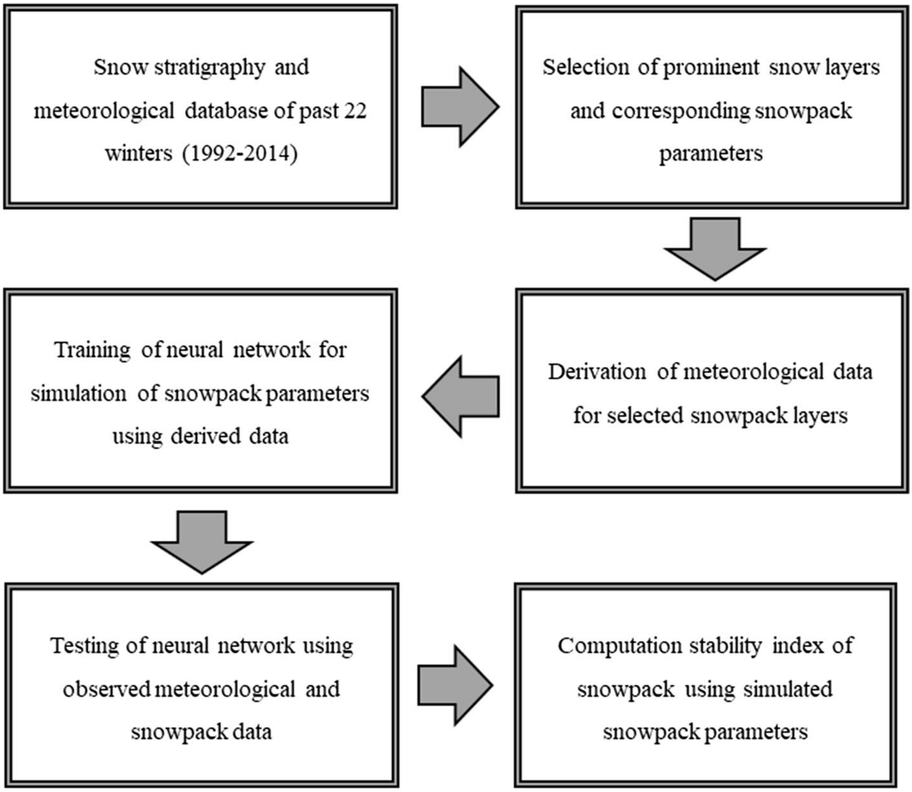

Fig. 2 Steps involved in the simulation of snowpack parameters and derivation of stability index of snowpack

The agreement scores of the individual parameters have been combined as weighted sum to yield the overall score of agreement between the observed and simulated profiles. In order to get an agreement score of a profile, we have used the weights of 0.5, 0.3 and 0.2 for RAM resistance, snow temperature and snow layer density, respectively.

The simulated snowpack parameters—RAM resistance, shear strength, density and thickness of snow layers—have been used to derive stability index of different snowpack layers. The stability index of the weakest snowpack layer (snowpack layer having lowest value of the RAM hardness and shear strength) has been defined as stability index of snowpack (SIS) as follows:

$$\begin{aligned} \text{Stability index of Snowpack (SIS)} &= (\text{Strength of the weakest snowpack layer})/(\text{overburden pressure}) \\ &= (\text{Normal strength i.e. RAM resistance (kg)} + \text{shear strength (kg)}) \\ &\quad /\left(\text{snow layer density } \left(\text{kg/m}^3\right) \times \text{snow volume above weak layer } \left(\text{m}^3\right)\right) \end{aligned}$$

The SIS and other snowpack parameters such as snowpack height, snow temperature of the weakest layer, snow load above the weakest layer and snowpack settlement have been used for prediction of the levels of avalanche danger and avalanche occurrence. In addition to the above snowpack data, avalanche warning bulletins (AWB) issued by SASE and record of avalanche occurrences have also been used. Snow and meteorological and

Table 2 Contingency matrix (4X4) of observed and forecasted events

Observed	Forecasted				
	No danger (N)	Low danger (L)	Medium danger (M)	High danger (H)	Total
No danger (N)	NN	NL	NM	NH	O_N
Low danger (L)	LN	LL	LM	LH	O_L
Medium danger (M)	MN	ML	MM	MH	O_M
High danger (H)	HN	HL	HM	HH	O_H
Total	F_N	F_L	F_M	F_H	Grand total (T)

Table 3 Contingency matrix (2X2) of observed and forecasted events

Observed	Forecasted		
	Avalanche occurrence (Y)	Non-occurrence (N)	Total
Avalanche occurrence (Y)	YY	YN	O_{YES}
Non-occurrence (N)	NY	NN	O_{NO}
Total	F_{YES}	F_{NO}	Grand total (T)

simulated snowpack data during winters from 1992 to 2014 have been used for development of ANN-based avalanche prediction model. During this period, there were total 8854 data points that include two observations daily at 0830 h and 1730 h. The model has been developed using data collected and derived for 0830 h only. The days having any of the model input variable missing have been removed from the database. This resulted in 2590 effective data points for training of the model. The training data set involves 427 avalanche days and 2163 non-avalanche days. The models predicting avalanche warnings and avalanche occurrences have been validated during winters 2014–2019. Contingency tables used for validation of these models are given in Tables 2 and 3. These tables have been used to compute various skill scores and accuracy measures such as True Skill Score (TSS), Heidke Skill Score (HSS), per cent correct (PC), bias, false alarm ratio (FAR) and probability of detection (POD). The models have been validated for avalanche occurrences during all danger levels and for individual avalanche danger levels as validated by Schirmer et al., 2009. The model validation scores have been computed as follows:

3.1 Accuracy measures and skill scores for 4 × 4 contingency matrix

A total number of observed No Danger events are given by:

$$O_N = NN + NL + NM + NH$$

A total number of forecasted No Danger events are given by:

$$F_N = NN + LN + MN + HN$$

Similarly, a total number of observed and forecasted events can be computed for low, medium and high danger from Table 2.

A total number of observed and forecasted events are given by:

$$T = \left(\mathrm{O_N} + \mathrm{O_L} + \mathrm{O_M} + \mathrm{O_H}\right) = \left(\mathrm{F_N} + \mathrm{F_L} + \mathrm{F_M} + \mathrm{F_H}\right)$$

The forecast verification measures derived from the above calculations are computed as follows:

$$\text{Percent Correct (PC)} = ((\mathrm{NN} + \mathrm{LL} + \mathrm{MM} + \mathrm{HH})/T)$$

$$\text{Heidke Skill Score (HSS)} = \left(\mathrm{NN} + \mathrm{LL} + \mathrm{MM} + \mathrm{HH} - \left(\mathrm{O_N} \times \mathrm{F_N} + \mathrm{O_L} \times \mathrm{F_L} + \mathrm{O_M} \times \mathrm{F_M} + \mathrm{O_H} \times \mathrm{F_H}\right)/T\right) \\ / \left(T - \left(\mathrm{O_N} \times F_\mathrm{N} + \mathrm{O_L} \times \mathrm{F_L} + \mathrm{O_M} \times \mathrm{F_M} + \mathrm{O_H} \times \mathrm{F_H}\right)/T\right).$$

3.2 Accuracy measures and skill scores for 2×2 contingency matrix

The forecast verification measures using 2×2 contingency matrix (Table 3) can be derived as follows:

$$\text{Percent correct (PC)} = \left((\mathrm{YY} + \mathrm{NN})/\mathrm{O_{YES}} + \mathrm{O_{NO}}\right) = \left((\mathrm{YY} + \mathrm{NN})/\mathrm{F_{YES}} + \mathrm{F_{NO}}\right)$$

$$\text{Bias} = \mathrm{F_{YES}}/\mathrm{O_{YES}}$$

$$\text{Probability of detection (POD)} = \mathrm{YY}/\mathrm{O_{YES}}$$

$$\text{False alarm ratio (FAR)} = \mathrm{NY}/\mathrm{F_{YES}}$$

$$\text{Heidke skill score (HSS)} = 2 \times (\mathrm{YY} \times \mathrm{NN} - \mathrm{YN} \times \mathrm{NY})/(\mathrm{YY} + \mathrm{YN}) \\ \times (\mathrm{YN} + \mathrm{NN}) + (\mathrm{YY} + \mathrm{NY}) \times (\mathrm{NY} + \mathrm{NN})$$

In neural network, the input neurons receive the model input data and process it according to the defined hyper-parameters for optimisation of weights according to the standard feed-forward neural network algorithm as discussed below:

3.3 Feed-forward neural network

Neural network model was first proposed by Frank Blosenblatt in 1958 (Harris, 2011). It consists of six basic components—inputs, weights, summing function, bias, activation function and output. The fundamental computational unit of the neural network is neuron or perceptron. A feed-forward neural network is formed of an input layer, one or more hidden layers, and an output layer of neurons. Each neuron in the input layer can receive a single input value. The neurons of hidden and output layer can accept an arbitrary number of inputs depending on the synapse type chosen to interconnect the neurons. Each neuron in the input and hidden layer is associated with a weight. In feed-forward network, there can

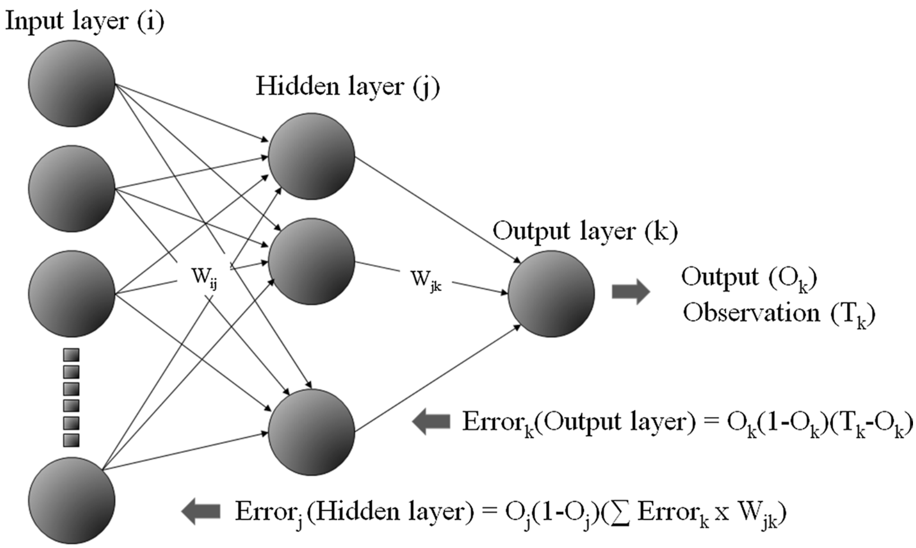

Fig. 3 Schematic of feed-forward neural network with back-propagation algorithm

be an arbitrary number of hidden layers, each of which may contain an arbitrary number of neurons. The schematic of a feed-forward neural network is shown in Fig. 3.

The neural network has an activation function that scales the output from a given layer to a useful output value (Heaton 2010). Selection of the activation function depends upon the desired numerical output for the application. In the present work, since input as well as output data have been normalized between 0 and 1, a sigmoidal activation function (Russell and Norvig 2010) is used. The activation function in the hidden layer performs a nonlinear operation on the output from the hidden neurons and serves it as an input to the neurons of the output layer. The activation function in the output layer gives the model output. At the output, error is computed as the difference of actual and modelled output and propagated back in the network to adjust the weights associated with each neuron to minimize the error. The back-propagation of error adjusts the weights to reach global minima of the error using gradient descent optimisation algorithm. The amount of weight adjusted with respect to the error function is known as learning rate or step size of the neural network. During back-propagation, the algorithm may stick in a local minima and may think that it had reached global minima thus producing sub-optimal results. To avoid this situation and to achieve optimal results, a momentum term is introduced to increase the step size of the algorithm so that it may jump out of the local minima. Achieving suitable value of learning rate and momentum for a neural network is a hit and trial process. The detailed mathematical formulation of the neural network has been expressed as follows:

The neural network first sums its weighted inputs X_1, X_2, X_3, … X_n and then passes the result through an activation function 'f' according to the following relation:

$$Y_j = f\left(\sum_{i=1}^{n} W_{ji} x X_i + \theta_j\right)$$

where Y_j represents output of the neuron, θ_j, an external threshold called bias and W_{ji}, the weight of the input X_i. Since the input and output have been normalised between zero and one, at each unit in the hidden and output layer a sigmoid activation function that gives output between zero and one has been used. The sigmoid function of a variable 'x' is defined by the following expression:

$$f(x) = 1/(1 + \exp(-x))$$

At the output layer, the error is computed using the following expression:

$$E_r = O_r \times \left(1 - O_r\right) \times \left(T_r - O_r\right)$$

where O_r is output of the activation function applied at the output layer, T_r is true value of the output and the product $O_r \times (1 - O_r)$ is rate of change of activation function.

The error computed at the output layer is propagated backward for upgradation of weights and biases. The error at the hidden layer is computed by the following expression:

$$E_h = O_h \times \left(1 - O_h\right) \times \left(\sum_r E_r \times W_{hr}\right)$$

In the nth iteration, the new weights and biases, W_{ij}' and θ_j', respectively, are computed by using the following expressions:

$$\Delta W_{ij}(n) = l \times E_j \times O_i + m \times \Delta W_{ij}(n-1)$$

$$W_{ij}' = W_{ij}' + \Delta W_{ij}$$

$$\Delta\theta_j = l \times E_j$$

$$\theta_j' = \theta_j + \Delta\theta_j$$

where 'l' and 'm' represent learning rate and momentum, respectively. During training of the neural network, different sets of learning rate and momentum between zero and one are used and retained the set producing minimum error. For better training of neural network, learning rate is kept smaller. However, smaller learning rate leads to longer time duration for training of the network. The learning rate and momentum are introduced for faster convergence of the network. The process of upgradation of weights is continued till a threshold error or number of epochs reached.

4 Results and discussion

Neural network-based model 'HIM-STRAT' has been developed for simulation of RAM hardness, shear strength, temperature, density, settlement of snowpack layers and prediction of avalanche danger and avalanche occurrence in Chowkibal–Tangdhar region of North-West Himalaya. The neural networks for all the snowpack parameters have been developed with 100,000 iterations and four sets of momentum and learning rates. The set of hyper-parameters that corresponds to the minimum error have been used for testing of the models. The hyper-parameters of the neural networks such as weights, biases,

Table 4 Hyper-parameters of neural network resulting smallest error of simulation of different snowpack parameters

Snowpack parameters	Learning rate	Momentum	No. of iterations	Initialisation range of weights and biases
1. RAM resistance	0.35	0.65	100,000	0.1– 0.001
2. Snow density	0.2	0.4	100,000	0.1– 0.005
3. Shear strength	0.4	0.5	100,000	0.1– 0.001
4. Snow temperature	0.25	0.35	100,000	0.1– 0.001
5. Snowpack settlement	0.3	0.5	50,000	0.1 – 0.01

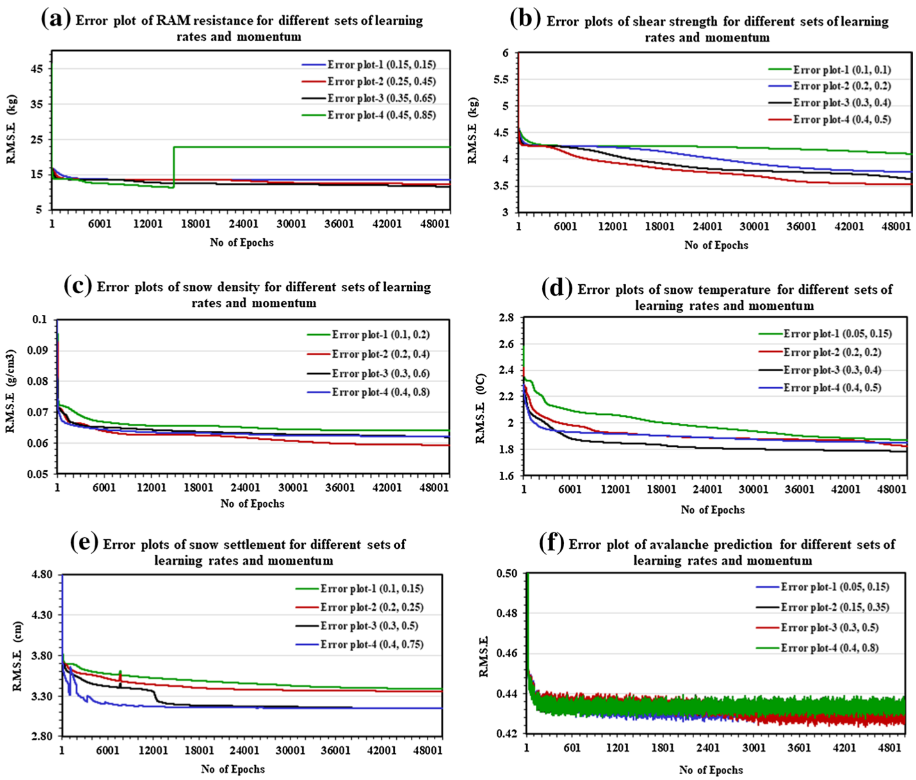

Fig. 4 **a–f** Plots of root-mean-square error (RMSE) of simulated snowpack parameters with No. of epochs for different sets of learning rates and momentum

momentum, learning rate and number of epochs that resulted in the smallest error are summarised in Table 4. The RMSE of neural network has been computed with 100,000 iterations and four different sets of learning rate and momentum. The plot of RMSE of snowpack parameters with the number of epochs for different sets of momentum and learning rates is shown in Fig. 4a–f. The RMSE of all the parameters for each set of learning rate and momentum decreases with the number of iterations except that for

RAM hardness. In the case of RAM hardness, for higher learning rate (0.45), the neural network initially converges rapidly up to approximately 15, 000 iterations beyond which it diverges and gives larger error. With higher learning rates though the neural network learning becomes faster, it can produce sub-optimal results or even diverge. The optimum results for all the models have been obtained with a set of learning rate and momentum in which learning rate was smaller than the momentum. The neural network models can be trained with other possible sets of the hyper-parameters and larger number of iterations to obtain global minima of the error. The accuracy of training of neural network models is highly dependent on the size of database. With larger database the network can be trained more precisely and can give better results. In this study, however, prominent snow layers observed on zero aspect have only been taken, thus forming a database of 613 data points. The database can further be enhanced by adding stratigraphy data of other regions having similar aspect and altitude within the same snow climatic zone.

The scatter plots of simulated and observed snowpack parameters for test data are shown in Fig. 5a–e. The ANNs for different snowpack parameters have been validated through computation of root-mean-square error (RMSE) of both training and testing data. The RMSE of simulated snowpack parameters for training data has been found smaller than that for the test data. However, the difference in RMSE of training and test data has not been found significant. The RMSE of RAM resistance, shear strength, snow density, snow temperature and snowpack settlement with test data has been found 9.2 kg, 3.4 kg, 0.05 g/cm^3, 1.8 °C and 3.7 cm, respectively. The comparison of RMSE and standard deviation of snowpack parameters for test data as shown in Fig. 6 a–e reveals that the RMSE of all the simulated parameters has been found reasonably smaller than the standard deviation of these parameters.

The HIM-STRAT has been used to simulate snowpack during two winters 2013–2014 and 2014–2015. The simulated snowpack during these winters has been compared with the observed snow profiles collected on zero aspect. The observed and simulated stratigraphy of zero aspect on 27 January 2014, 07 March 2014, 28 January 2015 and 23 February 2015 shown in Fig. 7a–d, respectively, has been compared quantitatively by computing an overall agreement score as proposed by Lehning et al. (2001). The weighted average of individual agreement scores for RAM resistance, snow temperature and snow layer density has been used to compute overall agreement score of the snowpack. The agreement scores for the above-mentioned four different snowpacks are summarised in Fig. 8. The analysis of agreement scores of the above snowpacks reveals that the RAM resistance and the snow density show a fairly good agreement score of about 0.8. The agreement scores for the snow temperature vary from 0.52 to 0.94. It has been observed that the snow temperature of a shallow snowpack as observed on 28 January 2015 has shown fairly good agreement score of 0.94. The overall agreement scores of the simulated snowpacks on the above-mentioned dates have been found 0.82, 0.78, 0.83 and 0.75, respectively. The qualitative assessment of all four snowpacks reveals that the simulated snowpack parameters—RAM resistance, snowpack height, snow temperature and snow layer density, considerably resemble with the corresponding parameters in the observed snow stratigraphy. Though in some of the simulated profiles it has been observed slight deviation in the number and thickness of snow layers from the observed ones, merging of snow layers of similar grain type, hardness and density in the observed snow stratigraphy results in the remarkable resemblance with the snowpack parameters of simulated snowpack.

Avalanche warnings and occurrences predicted by HIM-STRAT have been compared with the AWBs issued by SASE and avalanches observed during winters 2014–2019. The

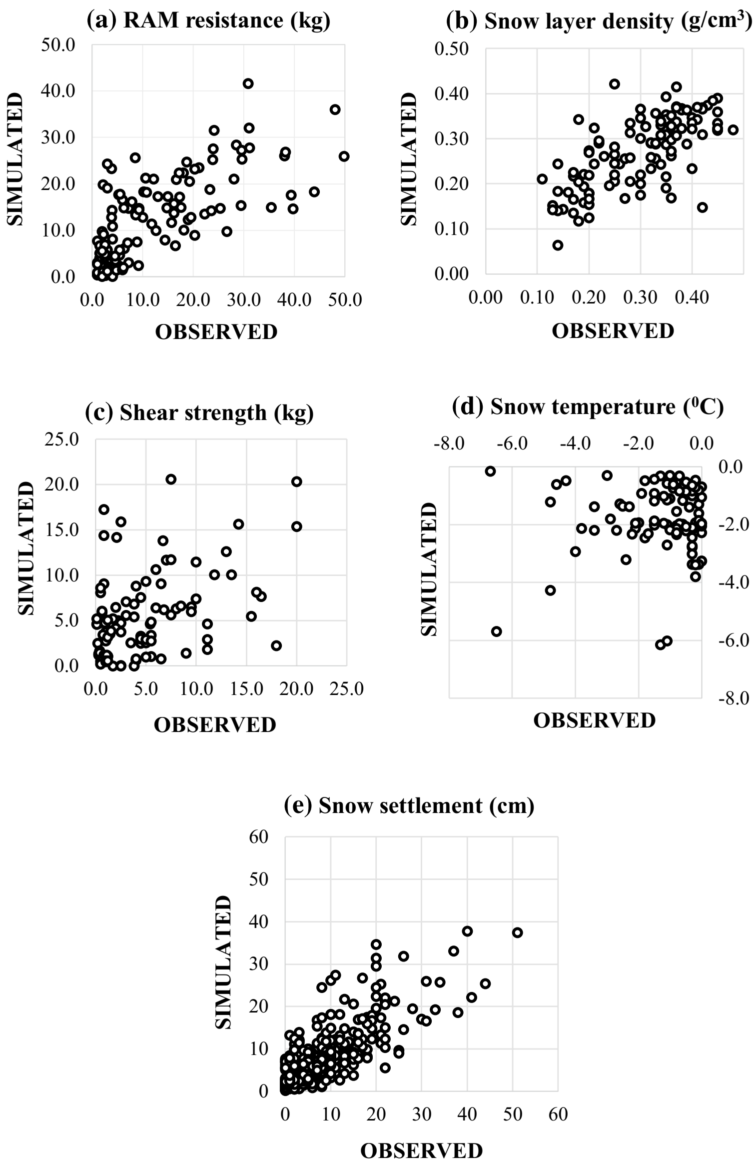

Fig. 5 **a–e** Scatter plot of simulated and observed snowpack parameters for test data

model predicting avalanche warnings has been validated for individual as well as combined avalanche danger as done by Schirmer et al. (2009). The performance of both the models has been assessed by computing HSS, TSS, PC, bias, FAR and POD. The contingency tables showing a number of avalanche warnings and occurrences during winters 2014–2019 are summarised in Tables 5 and 6 respectively.

During winters 2014–2019, total 606 days have been taken for validation of HIM-STRAT predictions for avalanche warnings and occurrences. Out of 606 days, official

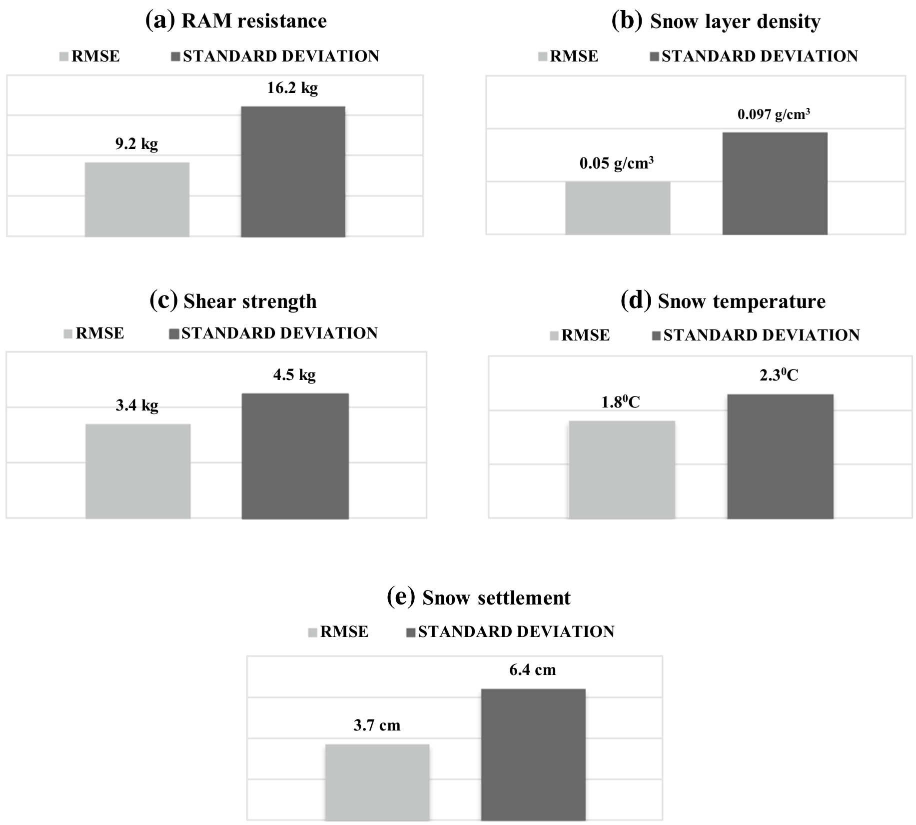

Fig. 6 **a**–**e** Comparison of root-mean-square error and standard deviation of simulated snowpack parameters

avalanche warning bulletin of SASE (AWB) issued no danger for 334 days, low danger for 118 days, medium danger for 122 days and high danger for 32 days. The HIM-STRAT predicted 361 days as no danger, 168 days as low danger, 56 days as medium danger and 21 days as high danger. Comparison of the model predictions with the AWBs issued by SASE resulted in the scores of PC, POD, FAR, bias, TSS and HSS for no danger as 0.83, 0.87, 0.20, 1.08, 0.65 and 0.66, for low danger as 0.74, 0.54, 0.62, 1.4, 0.33 and 0.28, for medium danger as 0.82, 0.34, 0.36, 0.53, 0.3 and 0.35 and for high danger as 0.97, 0.6, 0.1, 0.65, 0.6 and 0.7, respectively. The overall PC and the HSS of HIM-STRAT for prediction of avalanche warnings have been found 0.71 and 0.49, respectively. These scores signify that that the model performance is better for prediction of no and high danger than that for low and medium danger. When compared the results of HIM-STRAT with nearest neighbour model (KNN) presented by Schirmer et al. (2009), the TSS of HIM-STRAT for no (0.65) and high (0.60) danger cases and POD for avalanche predictions (0.77) have been found better than that of the KNN. As the performance of different models presented by Schirmer et al. (2009) is based only on the hit rate (POD) of forecast of different models for all danger levels and the TSS for individual danger levels, it is difficult to guess biases and false alarms of these models. It has generally been observed that a balance of hit rate, bias

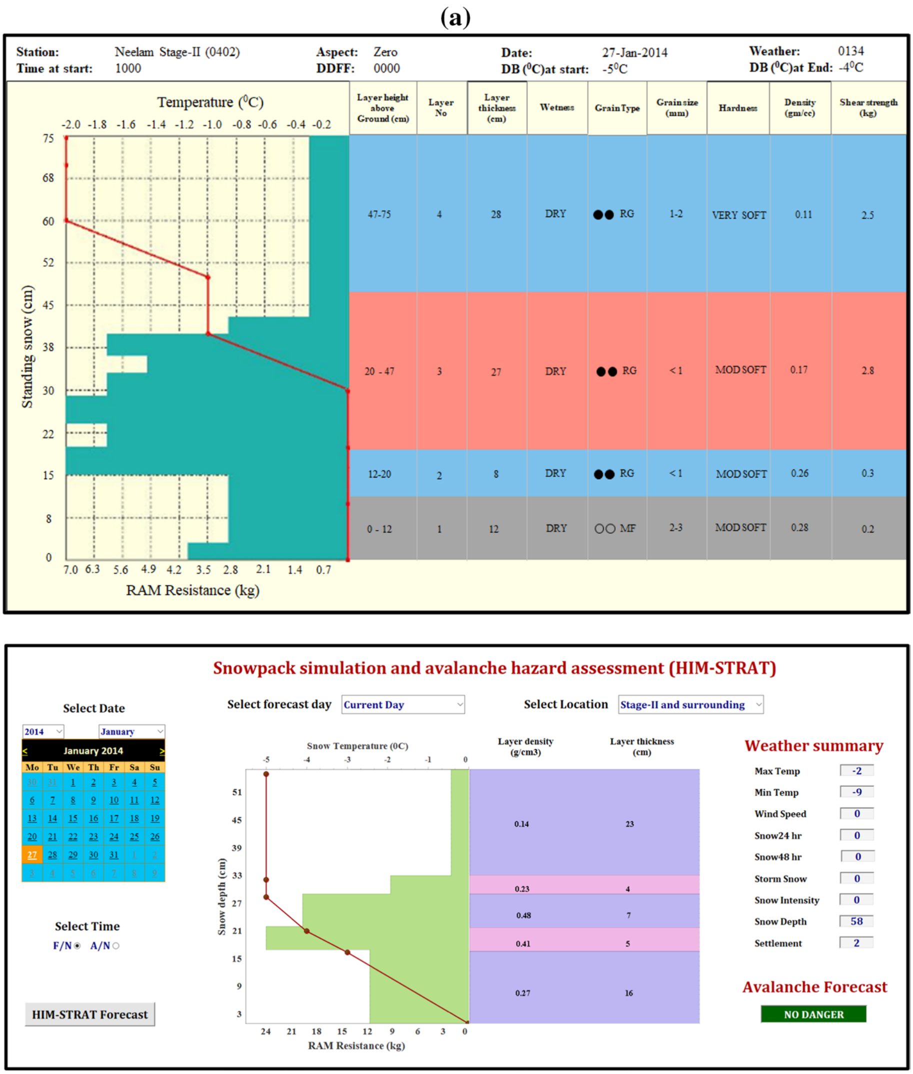

Fig. 7 Plots of (i) observed and (ii) simulated snow stratigraphy dated **a** 27 January 2014, **b** 07 March 2014, **c** 28 January 2015, **d** 23 February 2015

and false alarm is reflected in the HSS and its value greater than 0.25 for the forecast of a natural random phenomenon can be considered better than the random forecast.

In the C-T region during winters 2014–2019, out of total 606 days there were 100 avalanche days and 506 non-avalanche days. The avalanche occurrence data being skewed towards non-occurrence cases, only PC value of the model may not provide complete information for assessment of model performance. Hence, for assessment of performance of the HIM-STRAT for avalanche warning and occurrence predictions, in addition to the PC other accuracy measures and skill scores such as HSS, TSS, bias, FAR and POD have also been computed. Out of total 606 days, the HIM-STRAT predicted 245 days as

(b)

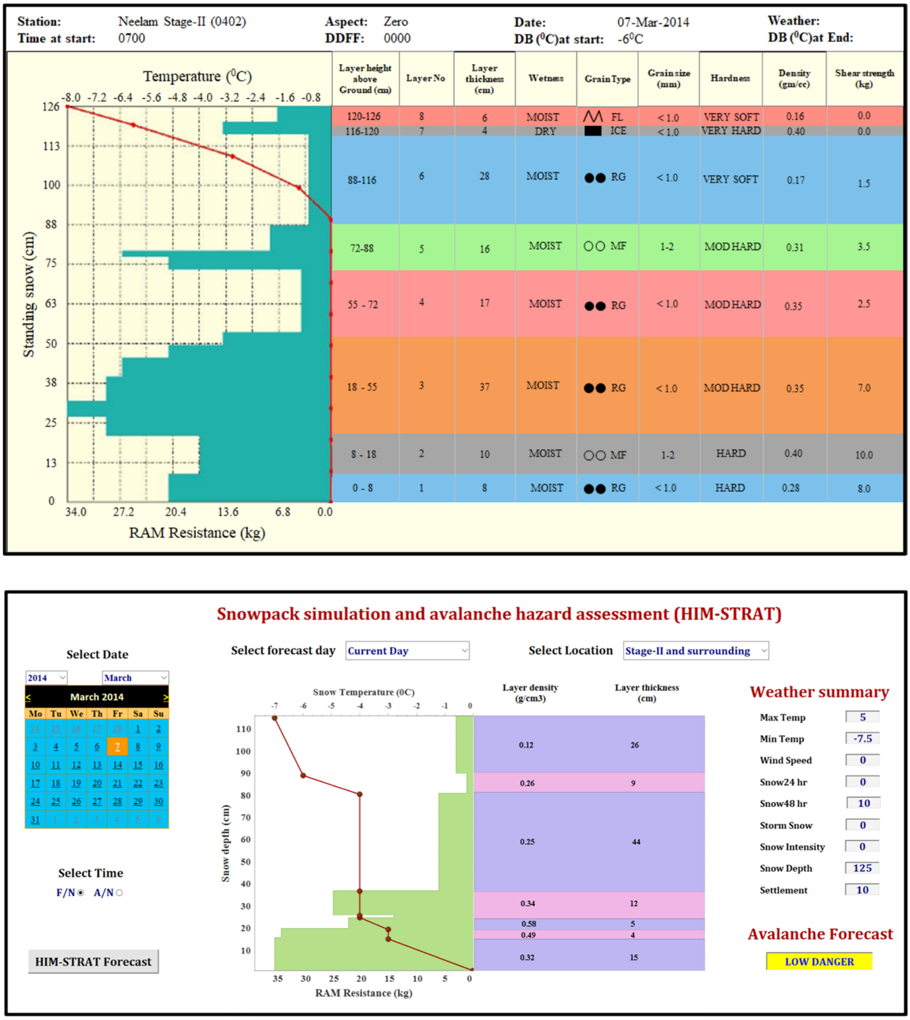

Fig. 7 (continued)

avalanche days and 361 days as non-avalanche days resulting in the POD—0.77, TSS—0.44, and HSS—0.28. The model performance scores for individual danger levels and avalanche occurrence are shown in Fig. 9. The POD of HIM-STRAT predictions for avalanche occurrences during all danger levels has been found better (0.77) than that of the KNN (0.73) presented by Schirmer et al. 2009. For some of the statistical avalanche forecasting models such as HMM (Joshi et al. 2018) though the PC (0.86) has been found better than that of the HIM-STRAT, overall performance measures such as bias, HSS and POD of HIM-STRAT have been found better. As the POD and the FAR are associated with type-1 error, higher POD and smaller FAR lead to smaller type-1 error. In avalanche work due to

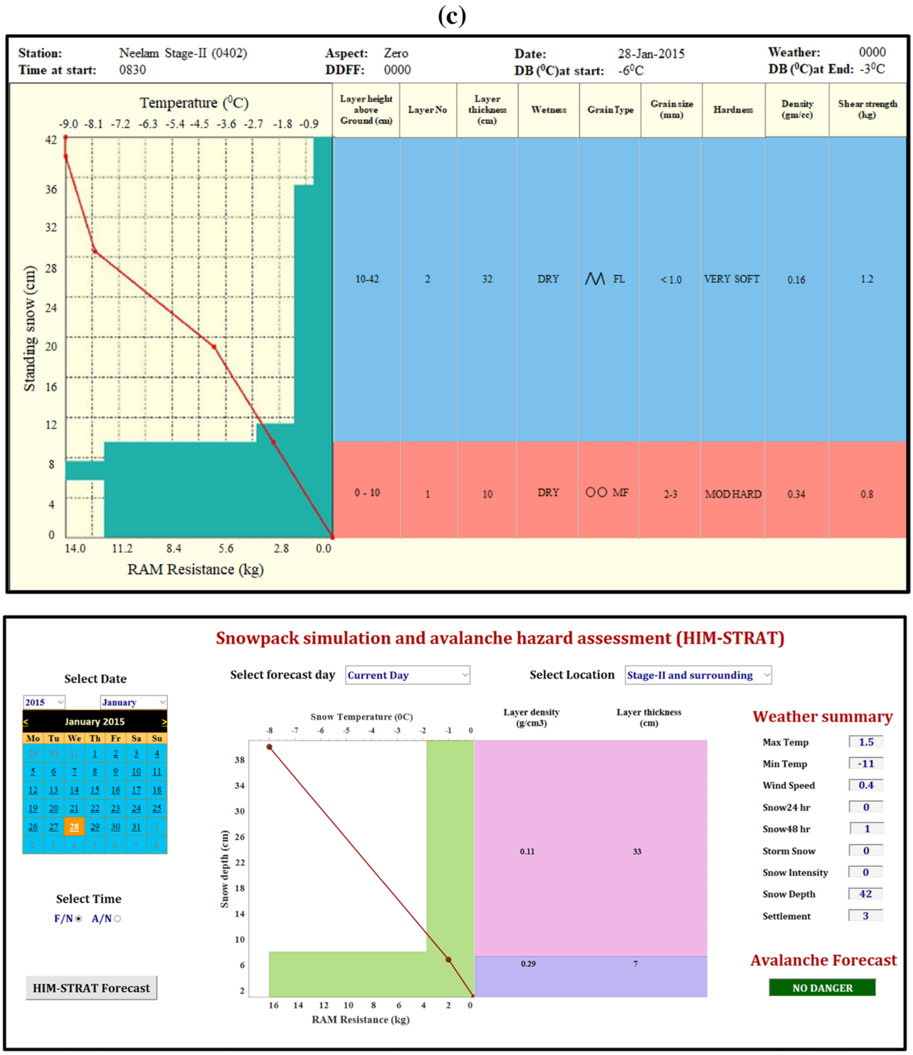

Fig. 7 (continued)

involvement of human lives, prediction of avalanche occurrence events is more important than that of non-occurrence events. Hence, a forecast with higher POD and smaller PC and FAR is considered better than that with smaller POD and higher PC and FAR. However, a balance between POD and FAR must be maintained for usable forecast.

In Pir Panjal and Great Himalayan ranges, most of the avalanches trigger during or immediate after snow storms and hence most of the avalanche forecasting models including HIM-STRAT have been developed for direct action avalanches. The stratigraphy database of late winter/spring season being limited for C-T region, the HIM-STRAT could not be developed

(d)

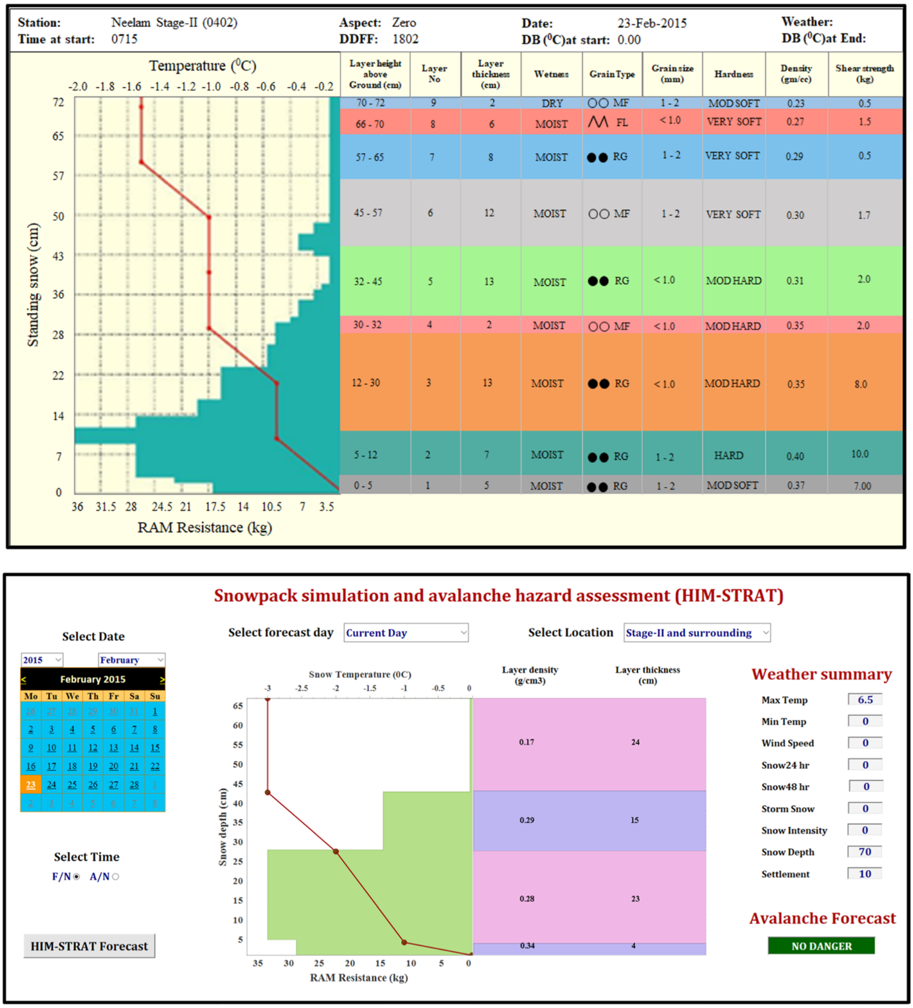

Fig. 7 (continued)

for simulation of late winter/spring snowpack and prediction of avalanches. However, stratigraphy data of late winter/spring season of other regions having similar climatic conditions as that of C-T region can be combined for simulation of isothermal snowpack parameters and prediction of late winter delayed/wet avalanches.

5 Conclusion

ANN-based snow cover simulation and avalanche prediction scheme HIM-STRAT simulates RAM resistance, shear strength, snow temperature, snow layer density and snowpack settlement with an RMSE of 9.2 kg, 3.4 kg, 1.8 °C, 0.05 g/cm^3 and 3.7 cm, respectively.

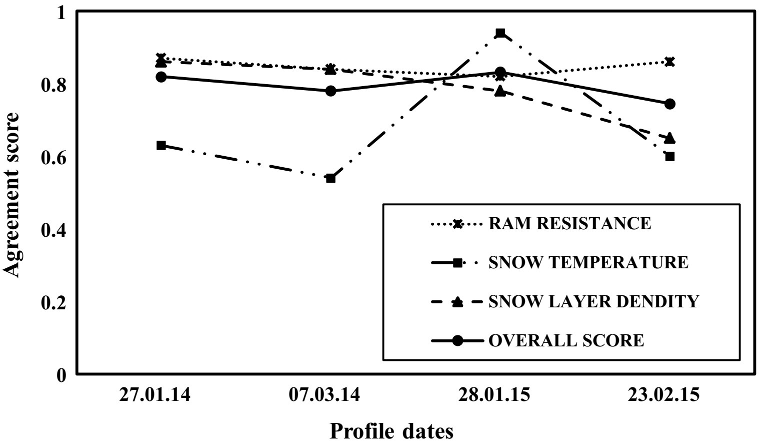

Fig. 8 Plot of agreement scores of simulated snowpack and snowpack parameters on different dates during winter 2013–2014 and 2014–2015

Table 5 Contingency table for validation of HIM-STRAT forecast against avalanche warning bulletin (AWB) issued during 05 winters (2014–2019)

Observed	Forecasted				
	No danger (N)	Low danger (L)	Medium danger (M)	High danger (H)	Total
No danger (N)	296	37	1	0	334
Low danger (L)	47	64	7	0	118
Medium danger (M)	18	61	41	2	122
High danger (H)	0	6	7	19	32
Total	361	168	56	21	606

Table 6 Contingency table for validation of HIM-STRAT forecast against actual avalanche occurrences during 05 winters (2014–2019)

Observed	Forecasted		
	Avalanche occurrence (Y)	Non-occurrence (N)	Total
Avalanche occurrence (Y)	77	168	245
Non-occurrence (N)	23	338	361
Total	100	506	606

The RMSE of all the simulated snowpack parameters has been found smaller than their standard deviation. Forecasting of different levels of avalanche danger has been carried out with POD 0.87 for no danger, 0.54 for low danger, 0.34 for medium danger and 0.60 for high danger and the HSS 0.66 for no danger, 0.28 for low danger, 0.35 for medium danger and 0.70 for high danger. The model predicted avalanche occurrence events with POD 0.77 and HSS 0.28. The performance of HIM-STRAT has been found better for prediction of no and high dangers than that for low and medium dangers. The snowpack parameters

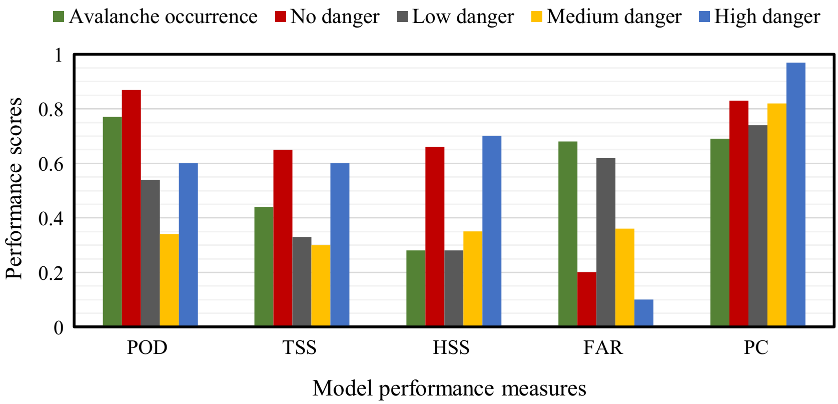

Fig. 9 HIM-STRAT performance measures for prediction of individual avalanche danger levels and avalanche occurrences during winters 2014–2019

simulated for the current day have been used for prediction of avalanche warnings and occurrences with a lead time of 24 h. The lead time of avalanche forecast can be increased by using forecasted weather data for simulation of future snowpack. Since the performance of avalanche forecasting models is dependent on simulated snowpack parameters, accuracy in simulation of snowpack parameters will lead to increase the accuracy of avalanche forecast. The accuracy of snowpack parameters can further be improved by enhancing the training database and developing more optimised model by choosing optimal set of hyper-parameters and robust optimisation techniques.

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